

MACHINE LEARNING IN PHYSICS FOUNDATIONS 2

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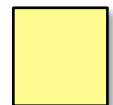
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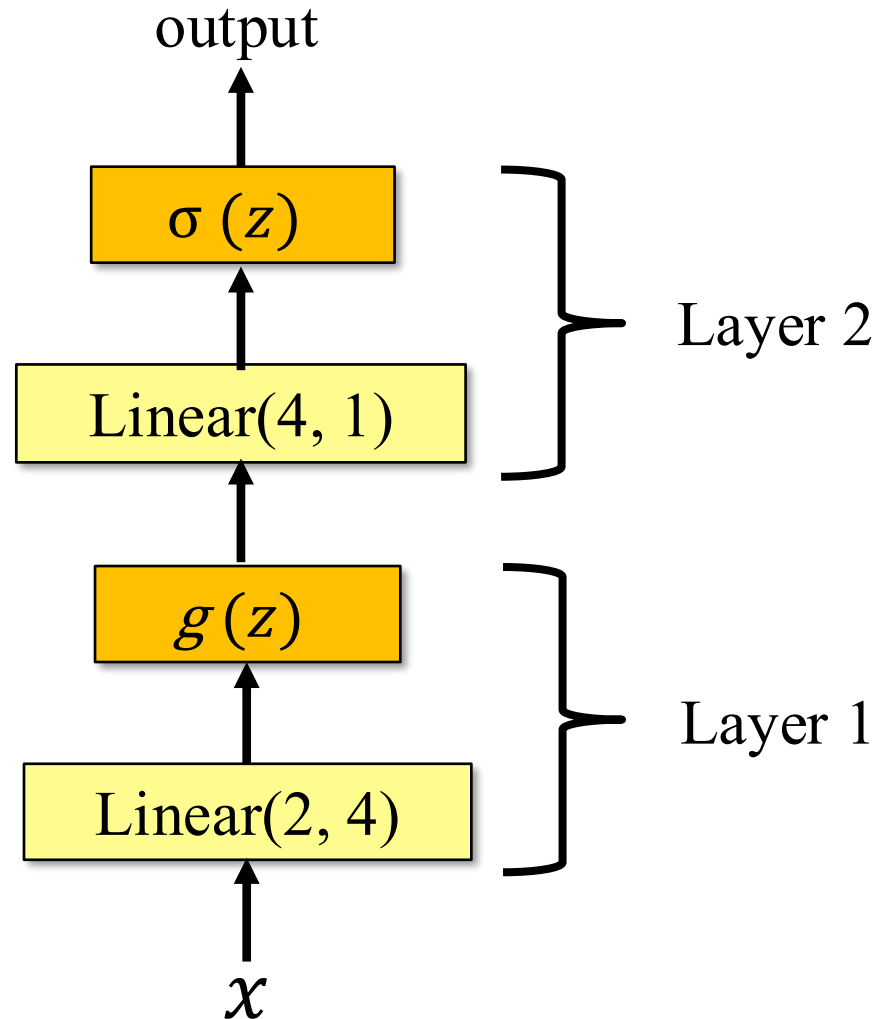
Recap

Last time we introduced the multi-node, multi-layer perceptron (MLP). This is also known as a **feed-forward neural network** (FCNN)

 $\sigma(z) = (1 + e^{-z})^{-1}$

 $g(z) = \text{relu}(z)$

 $f(z) = zA^T + b$



FITTING MODELS TO DATA

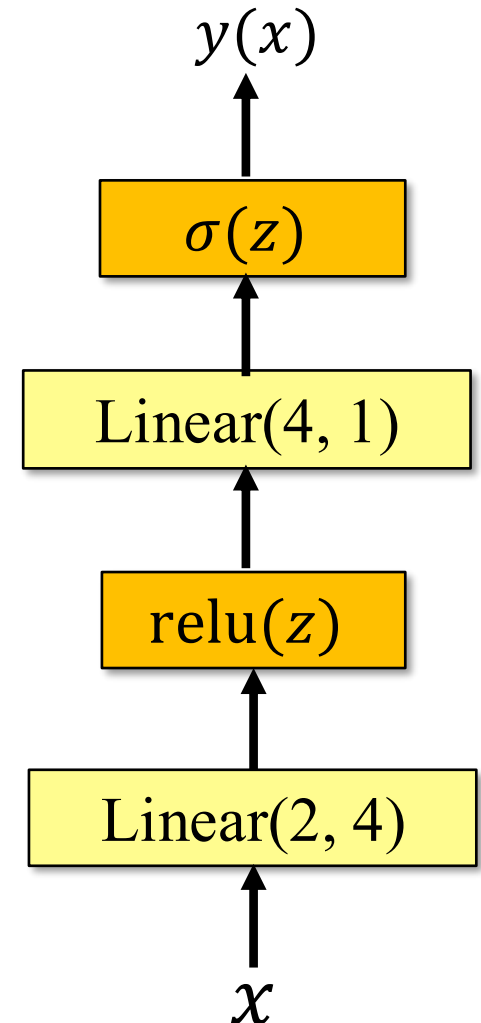
Fitting Models to Data (1)

Suppose our goal is to approximate an unknown function

$$y = f(x_1, x_2)$$

given a dataset $D = \{(x_1, x_2, y)_{i=1}^N\}$.

The standard approach is to approximate $f(x_1, x_2)$ with a *parameterized* function $f_{\omega}(x_1, x_2)$, that is, a model, where ω are the parameters of the model.



Fitting Models to Data (2)

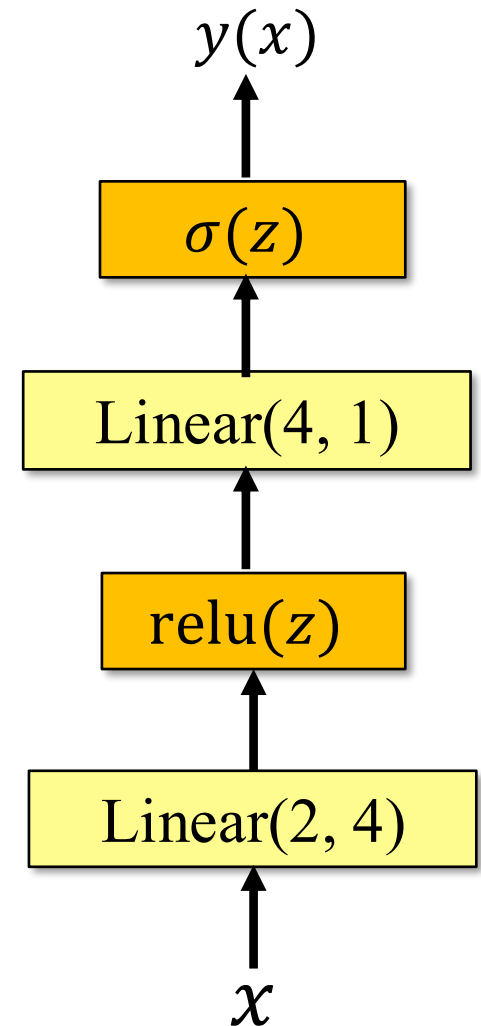
One way to find the *best-fit values* ω^* of the model $f_{\omega}(x_1, x_2)$ is to minimize the **quadratic form**

$$R(\omega) = \frac{1}{N} \sum_{i=1}^N (y_i - f_i)^2$$

with respect to the parameters ω , where

$$f_i \equiv f_{\omega}(x_{1i}, x_{2i}).$$

This is called a **least-squares fit**.



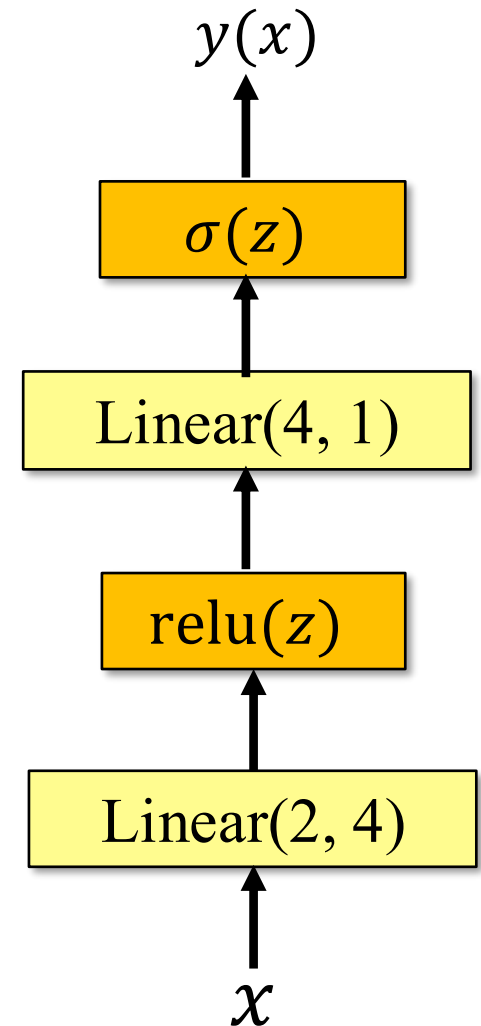
Fitting Models to Data (3)

The least-squares fit can be generalized to

$$R(\omega) = \frac{1}{N} \sum_{i=1}^N L(y_i, f_i)$$

where, $L(y_i, f_i)$, the **loss function**, measures the discrepancy between the desired **target** y and the model, f .

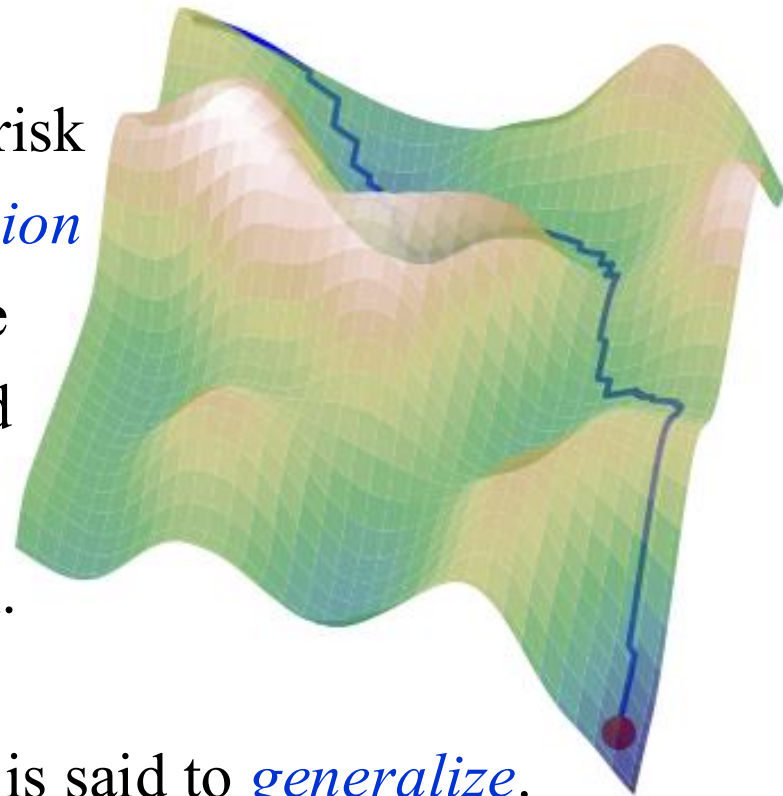
$R(\omega)$ is called the **empirical risk** or **average loss**. **Warning:** In ML, the empirical risk is often referred to as the “loss”.



Empirical Risk Landscape (1)

The empirical risk defines a highly corrugated high-dimensional “landscape” in the space of parameters.

Our task is to navigate the empirical risk landscape to find a *good approximation* of the minimum not of *that* landscape but rather of the landscape that would be formed if we had an infinite amount of data and the correct model.

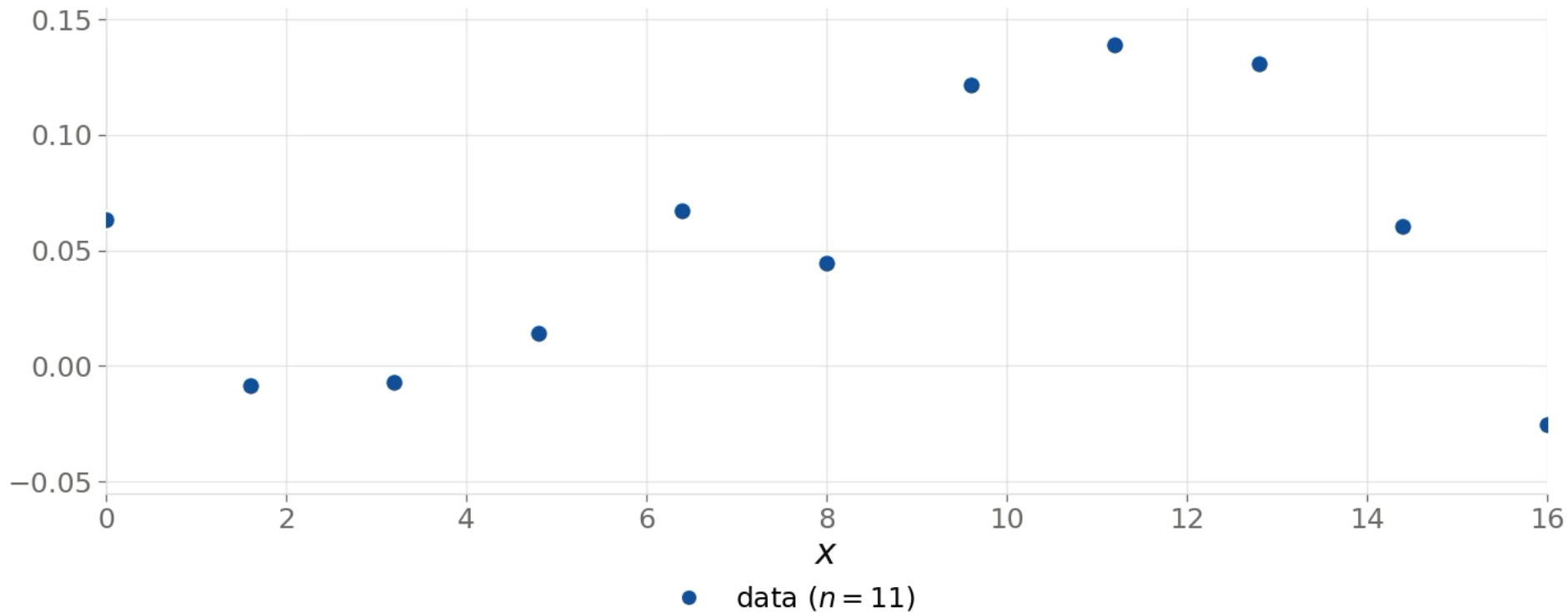


A model that approximates this ideal is said to *generalize*.

Empirical risk is what we can measure — generalization is what we want

We only ever see a finite, noisy sample.

Everything we compute is built from these 11 points.



Example: A least squares fit to a **linear model**.

$$f(x, \omega) = \sum_{j=0}^n \omega_j g_j(x), \quad g_j(x) = x^j$$

RISK FUNCTIONAL

Risk Functional (1)

In mathematics and physics, insight can often be gained by taking some limit of an expression. Consider the limit of

$$R(\omega) = \frac{1}{N} \sum_{i=1}^N L(y_i, f_i)$$

as $N \rightarrow \infty$, that is, as the amount of data grows without limit.

In that limit, the empirical risk becomes the **risk functional**,

$$R[f] = \int dx \int dy L(y, f) p(x, y)$$

where $p(x, y)dx dy$ is the probability distribution of the data.

Risk Functional (2)

To find the optimal function $f = f^*$, ideally, we should minimize the risk functional

$$R[f] = \int dx \int dy L(y, f) p(x, y)$$

While we know the functional form of the loss function $L(y, f)$ because we choose it, the probability distribution, $p(x, y)dx dy$, of the data is usually unknown.

However, $R(\omega)$ is a **Monte Carlo** approximation of $R[f]$, so any useful result derived for the risk functional pertains, at least approximately, to the empirical risk.

Risk Functional (3)

To minimize

$$R[f] = \int dx \int dy L(y, f) p(x, y)$$

first note that $p(x, y) = p(y|x) p(x)$.

Therefore, we can write the functional $R[f]$ as

$$R[f] = \int dx p(x) \mathcal{L}(x, f)$$

where,

$$\mathcal{L}(x, f) = \int dy L(y, f) p(y|x)$$

Risk Functional (4)

Add an arbitrary function $\epsilon g(x)$ to the optimal function f^* , where ϵ is an infinitesimal*.

$$\begin{aligned} R[f^* + \epsilon g] &= \int dx \, p(x) \, \mathcal{L}(x, f^* + \epsilon g) \\ &= \int dx \, p(x) \left(\mathcal{L}(x, f^*) + \epsilon g \frac{\partial \mathcal{L}}{\partial f^*} \right) \\ &= R[f^*] + \epsilon \int dx \, p(x) g(x) \frac{\partial \mathcal{L}}{\partial f^*} \end{aligned}$$

* $0 < |\epsilon| < |x|$, where x is any real number.

Risk Functional (5)

Rearranging, and taking the standard part* of both sides, we find

$$\text{st} \left(\frac{R[f^* + \epsilon g] - R[f^*]}{\epsilon} \right) = \int dx \, p(x) g(x) \frac{\partial \mathcal{L}}{\partial f^*}$$

The lefthand side *defines* the **functional derivative** $\delta R / \delta f$.

By assumption, $\delta R / \delta f$ is zero at $f = f^*$, $\forall x$. Therefore,

$$\int dx \, p(x) g(x) \frac{\partial \mathcal{L}}{\partial f} = 0$$

when $f = f^*$.

*The standard part of the **hyperreal** $x^* = x + \epsilon a$ is defined by $x = \text{st}(x^*)$

Risk Functional (6)

We want the expression

$$\int dx \, p(x) g(x) \frac{\partial \mathcal{L}}{\partial f} = 0$$

to hold for *any* function $g(x)$ and $\forall x$.

This can happen *if only if*

$$\frac{\partial \mathcal{L}}{\partial f} = 0$$

that is,

$$\frac{\partial \int dy \, L(y, f) p(y|x)}{\partial f} = 0$$

Risk Functional (7)

If $\partial/\partial f$ and $\int$ commute, we obtain the very important result:

$$\int dy \frac{\partial L}{\partial f} p(y | x) = 0 \rightarrow \mathbb{E}_{y \sim p(y|x)} \left[\frac{\partial L}{\partial f} \right] = 0$$

1. The result is *independent* of the details of $f_\omega(x)$.
However,...
2. The function $f_\omega(x)$ must have sufficient **capacity**: i.e., there must exist an approximation $f_{\hat{\omega}}(x)$ found by minimizing the empirical risk $R(\omega)$ that is close (in some precise sense) to the optimal function f^* .
3. Moreover, it must be possible to find that function!

COMMON LOSS FUNCTIONS

Common Loss Functions (1)

Quadratic loss (often used for regression)

$$L(y, f) = (y - f)^2$$
$$y \in \mathbb{R}$$

Binary cross entropy (often used for classification)

$$L(y, f) = -[y \log f + (1 - y) \log(1 - f)]$$
$$y \in \{0, 1\}$$

Quadratic Loss

Quadratic loss: $L(y, f) = (y - f)^2$

$$\mathbb{E}_{y \sim p(y|x)} \left[\frac{\partial L}{\partial f} \right] = 0$$

Solution

$$f(x, \omega^*) = \mathbb{E}_{y \sim p(y|x)} [y]$$

Very Important Point (VIP): As noted, the result is independent of the details of f . The result depends solely on the form of the loss function and the conditional probability, $p(y|x)dy$, of the data.

Binary Cross Entropy Loss (1)

Binary cross entropy loss : $-[y \log f + (1 - y) \log(1 - f)]$

$$\mathbb{E}_{y \sim p(y|x)} \left[\frac{\partial L}{\partial f} \right] = 0$$

$$\mathbb{E}_{y \sim p(y|x)} \left[\frac{y}{f} - \frac{(1 - y)}{1 - f} \right] = 0$$

Therefore, for y in the unit interval

$$f(x, \omega^*) = \mathbb{E}_{y \sim p(y|x)} [y]$$

But when $y = \{0, 1\}$ this yields

$$f(x, \omega^*) = p(1 | x)$$

Binary Cross Entropy Loss (2)

According to Bayes' theorem

$$f(x, \omega^*) = p(1|x) = \frac{p(x|1) \pi(1)}{p(x)}$$

with $p(x) = p(x|1) \pi(1) + p(x|0) \pi(0)$.

- $p(x|k)$ is the probability density of x for the class labeled $y = k$.
- $\pi(k)$ is the prior probability of the class labeled $y = k$.

Any model trained with binary cross entropy necessarily approximates $p(1|x)$.

Other Loss Functions

Exponential loss

$$L(y, f) = \exp\left(-\frac{wyf}{2}\right), \quad y \in \{-1, +1\}$$

Quantile loss ($0 \leq \tau \leq 1$)

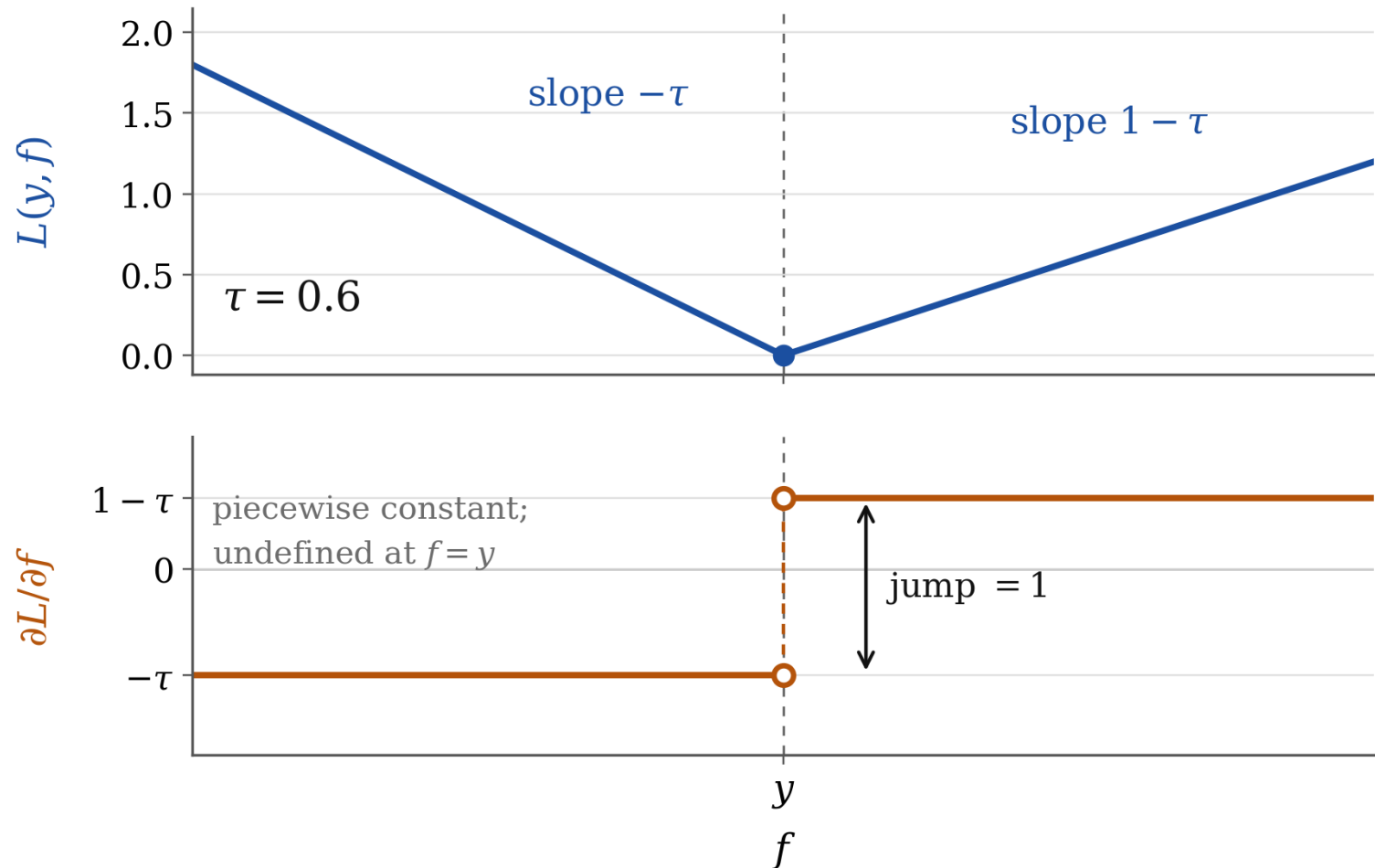
$$L(y, f) = \begin{cases} \tau(y - f) & y \geq f \\ (1 - \tau)(f - y) & y < f \end{cases}$$

Zero-One loss

$$L(y, f) = \begin{cases} 0 & |y - f| \leq b \\ 1 & |y - f| > b \end{cases}$$

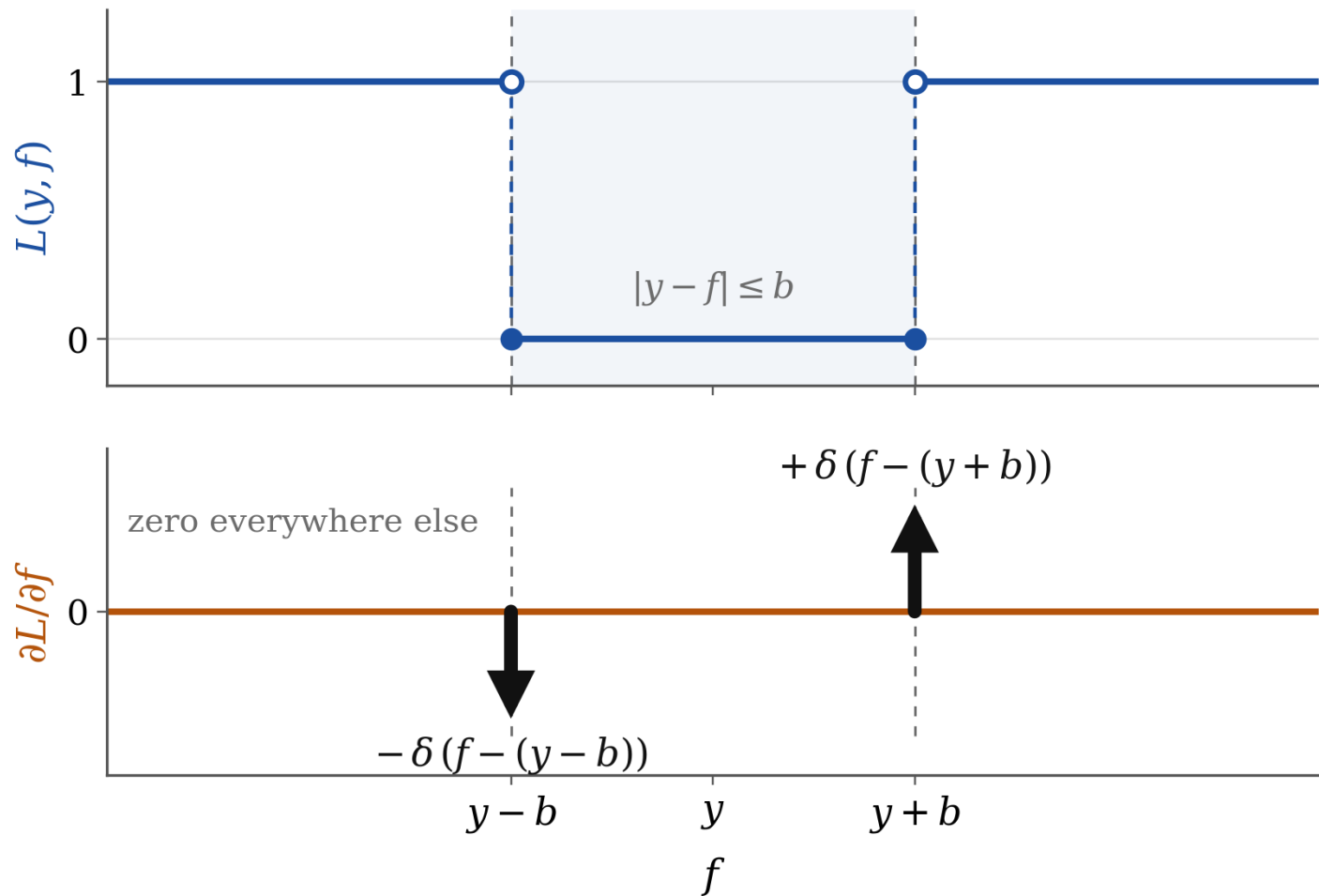
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Summary

Stochastic Gradient Descent

- Many minimization algorithms are variations of the following algorithm:

$$\omega_{j+1} = \omega_j - \eta \nabla R$$

- Batches of data are used, which introduces noise into ∇R .

Risk Minimization

- Given a data set $D = \{(x, y)\}_{i=1}^N$, a model $f_{\omega}(x)$, and a loss function $L(y, f)$, the optimal function $f_{\hat{\omega}}(x)$ satisfies:

$$\int \frac{\partial L}{\partial f} p(y | x) dy = 0$$

approximately.